🚀 How Do You Get Databricks Genie One/Agent Production-Ready?

📋 12 Simple Steps for Agent Routing & Quality

Getting Databricks Genie One production-ready comes down to two main goals: ensuring questions route to the right Genie Agent and making sure each agent returns highly accurate answers.

This guide has a breakdown on how Genie One routing works under the hood, what the Genie Ontology is, and 12 prioritized steps (ordered by impact and effort) to get your agents enterprise-ready.

🔀 How Does Genie One Route Questions?

When a user asks a question in Genie One chat, the system follows a clear three-step sequence:

1. **Search available Genie Agents** — Genie One scans accessible agents, matching relevance against agent names, descriptions, table metadata, instructions, and curated context.
2. **Route to match** — If a relevant agent is found, the prompt is routed to generate SQL against curated metric views or tables and return an immediate response.
3. **Fallback search** — If no agent matches, Genie One searches dashboards, queries, metric views, and inferred context (the Genie Ontology) to construct an answer.

📖 **Reference Docs:** AWS | GCP | Azure

🧠 What Is the Genie Ontology and Why Does It Matter?

Genie One relies on a unified context layer called the **Genie Ontology**, combining two distinct layers of context:

Layer	Description	Managed By
Unity Catalog Semantics (modeled)	Metric Views, Domains, Pages, Certification	You - governed explicitly
Inferred Context (automatic)	Snippets auto-extracted from dashboards, SQL queries, Genie Agents, and usage patterns. Each snippet gets an authority score based on source, usage frequency, and freshness.	Genie - maintained automatically

Inferred snippets dynamically extract metric definitions, source pointers, and critical business rules. All snippets remain strictly gated by Unity Catalog permissions.

📖 **Reference Docs:** AWS | GCP | Azure

🗺️ What Are the 12 Steps to Production-Ready Genie Agents?

Below is a roadmap ordered by effort and impact tier. Focus on the top items first to maximize impact with minimal effort.

💡 **Tip:** Many of these steps can be accelerated using Genie Code, which assists with agent creation, metadata suggestions, Knowledge Store curation, and benchmark analysis.

#	Action	Impact	Effort	Helps With
1	Focused, domain-specific agents	⭐⭐⭐⭐⭐ High	⭐ Low	Routing + Quality
2	Table and column metadata	⭐⭐⭐⭐⭐ High	⭐ Low	Routing + Quality
3	Agent names and descriptions	⭐⭐⭐⭐ High	⭐ Low	Routing
4	Permission-based visibility	⭐⭐⭐⭐ High	⭐ Low	Routing
5	Certification and deprecation	⭐⭐⭐⭐ High	⭐ Low	Routing
6	Knowledge Store	⭐⭐⭐⭐⭐ High	⭐⭐ Medium	Routing + Quality
7	Test, benchmark, and iterate	⭐⭐⭐⭐ High	⭐⭐ Medium	Routing + Quality
8	Metric views	⭐⭐⭐⭐⭐ High	⭐⭐⭐ Medium	Routing + Quality
9	Domains and subdomains	⭐⭐⭐ Medium	⭐ Low	Routing
10	Agent text instructions	⭐⭐⭐ Medium	⭐ Low	Quality
11	Pages (glossary)	⭐⭐⭐ Medium	⭐⭐ Medium	Routing + Quality
12	Workspace instructions	⭐⭐ Moderate	⭐⭐ Medium	Routing

🛠️ How Do You Implement Each Step?

🎯 1. Keep Agents Focused and Domain-Specific

One agent per business domain, ≤5 data objects each (max 50). Pre-join related tables into views or metric views (see step 8) to simplify the data model. Minimize table overlap between agents - shared tables confuse routing. 📖 **Reference Docs:** AWS | GCP | Azure

🏷️ 2. Table and Column Metadata

Write precise UC table and column descriptions. Genie uses these for both routing (finding the right agent) and SQL generation. Avoid vague or redundant descriptions.

📖 **Reference Docs:** AWS | GCP | Azure

✏️ 3. Write Clear Agent Names and Descriptions

Use descriptive names (*"Marketing Campaign Performance"* not *"Agent 1"*) and rich descriptions explaining what questions the agent answers. Match the terminology your users actually use.

📖 **Reference Docs:** AWS | GCP | Azure

🔒 4. Control Agent Visibility via Permissions

Share each agent only with the users/groups who need it (CAN VIEW minimum). This is the **primary lever** for controlling routing scope — there is no explicit allowlist in Genie One today.

📖 **Reference Docs:** AWS | GCP | Azure

🏅 5. Certify Production-Ready Agents

Mark validated agents as **Certified** using the UC `system.certification_status` tag. Mark outdated agents as **Deprecated**. Certification steers Genie One toward trusted assets when multiple could match.

📖 **Reference Docs:** AWS | GCP | Azure

📚 6. Build a Rich Knowledge Store

Add example SQL queries (most effective tuning tool), SQL expressions for business logic, join relationships, and prompt matching rules. Prioritize SQL-based definitions over plain text.

📖 **Reference Docs:** AWS | GCP | Azure

🧪 7. Test, Benchmark, and Iterate

Be your agent's first user. Create benchmark questions covering your most frequent queries with 2–4 phrasings each.

A complete benchmark set includes straightforward questions, edge cases, out-of-scope guardrails, and ambiguous queries. Use Genie Code for automated scoring and optimization.

📖 **Reference Docs:** AWS | GCP | Azure

📊 8. Define Canonical Metrics with Metric Views

Define KPIs as governed Metric Views (e.g., *"revenue = SUM(amount) / COUNT(DISTINCT customer_id)"*). These feed into the Genie Ontology, helping both routing and consistent metric resolution.

📖 **Reference Docs:** AWS | GCP | Azure

📂 9. Organize Assets into Domains

Group assets by business purpose using UC Domains. Domains appear in the Genie One sidebar and organize Pages by business area, improving discoverability and routing context.

📖 **Reference Docs:** AWS | GCP | Azure

💬 10. Agent Text Instructions (Sparingly)

Use as a last resort when SQL expressions and examples can't address the need. Be specific (*"When users ask about sales without specifying product, ask them to clarify"*), keep the set small, and avoid conflicts with SQL-based definitions. 📖 **Reference Docs:** AWS | GCP | Azure

📖 11. Define Business Terms in Pages

Author governed Pages for key terms (*"active user," "DAU," "qualified lead"*). Genie references these as the authoritative source for disambiguation in both routing and answer generation. If you already have business terms defined in Confluence or another knowledge repository, use Bulk Import Pages to bring them into Unity Catalog at scale.

📖 **Reference Docs:** AWS | GCP | Azure

⚙️ 12. Add Workspace Instructions for Global Routing Guidance

Workspace admins can create `/Workspace/.genie_workspace_instructions.md` to provide global context to Genie One chat — data conventions, preferred terminology, and routing directives (e.g., *"Marketing campaign questions should use the Marketing Campaign Performance agent"*). Write in plain English, one directive per bullet. Applies to Genie One chat only.

📖 **Reference Docs:** AWS | GCP | Azure

📌 Key Takeaways

- **Genie One routes by relevance**, not by a fixed priority list — invest in the signals that make your agent the most relevant match.
- **Start with the low-effort, high-impact wins**: focused agents, good metadata, clear names, permissions, and certification. These five steps cost almost nothing and dramatically improve routing.
- **The Knowledge Store is your most powerful tuning lever** — example SQL queries, SQL expressions, and join relationships improve both routing and answer quality.
- **Metric Views and Pages feed the Genie Ontology** — define your KPIs and business terms once, and Genie One uses them everywhere.
- **Benchmark before you ship** — test with multiple phrasings, include edge cases and out-of-scope questions, and aim for high accuracy on your most critical queries.
- **Use Genie Code** to accelerate agent creation, metadata curation, and benchmark analysis.

❓ FAQ

Can you train Genie One directly? No. You train the Genie Agents it routes to — using benchmarks, Knowledge Store, and metadata. The better curated your agents, the smarter Genie One's routing becomes.

Does certification guarantee routing priority? Certification is a trust signal that steers Genie One toward certified assets, but it's not a hard filter. Combine it with deprecating outdated assets and workspace instructions for the best results.

How many benchmark questions should an agent have? There's no fixed minimum. Cover your most frequently asked questions with 2–4 phrasings each, and include a mix of straightforward, edge-case, out-of-scope, and ambiguous questions.